

# Tat-Chuyen Mai

AI Researcher — Specializing in Computer Vision and Deep Learning

 LinkedIn  MT-Chuyen  gochuyen00@gmail.com  0325510961  Hanoi, Vietnam

## Career Objective

Dedicated AI researcher with **three years** of experience in developing & optimizing deep learning architectures for 1D & 2D signal processing, **first authored of 6 scientific papers** with **two Q1 Under Review Papers**. Strong background in computer vision, with hands-on experience in coding, training, and fine-tuning deep learning models using Python. Eager to apply vision-based AI to automation, building reliable models that bridge perception and intelligent decision-making.

## Education

**Hanoi University of Science and Technology (HUST)**

*Sep 2021 – Sep 2025*

Bachelor of Advanced Program in Biomedical Engineering

- GPA: **3.72/4.00**
- Academic Excellence Scholarship [🔗](#)
- Best Graduation Project [🔗](#)
- 3rd Prize (University) [🔗](#) – HUST Annual Student Research Conference 2024–2025
- Related subjects: Programming Languages, Probability and Statistics, Machine learning, Image processing

## Recent Under Review research

**A study on methods for segmenting medical images**

*Tat-Chuyen Mai*, Viet-Dung Nguyen\* - [BMC Bioinformatics \(Q1\)](#) [🔗](#)

- Proposed a dynamic adaptation method to address the problem of object size diversity.
- Developed a loss function that encourages learning multi-scale features

**A study on a recommendation model for predicting miRNA targeting**

*Tat-Chuyen Mai*, Nui-Nguyen Van, Duc-Hau Le\* - [Computer Methods and Programs in Biomedicine \(Q1\)](#) [🔗](#)

- Developed methods to solve the problem of data sparsity.

## Publications

- **Inception Vision Mamba Unet for Tumor Segmentation on Breast Ultrasound Images**

*Tat-Chuyen Mai*, Viet-Dung Nguyen\* - [Springer Nature](#) [🔗](#)

*Developed the Vision Mamba U-Net architecture integrating multi-scale feature learning from Inception modules, achieving a 3% improvement in segmentation accuracy on breast ultrasound datasets.*

- **Multichannel Learning Framework for Enhanced ECG Signal Classification**

*Tat-Chuyen Mai*, Diem-Anh Nguyen Thi, Quang-Huy Hoang, Viet-Huong Pham Thi, Anh-Vu Tran\* - [Iranian Journal of Science and Technology \(Q2\)](#) [🔗](#)

*Proposed a multimodal deep learning framework that integrates spectrogram images of ECG signals with wavelet-transformed filtered raw signals.*

- **Multimodal Deep Learning for Breast Ultrasound Lesion Classification**

*Tat-Chuyen Mai*, Khanh-Ly Trinh, Huyen-Trang Ong, Hai-Long Nguyen, Viet-Dung Nguyen\* - [ICHST 2025](#) [🔗](#)

*Proposed a multimodal deep learning framework integrating image and metadata representations to enhance classification of breast ultrasound lesions.*

- **Motion Analysis & Performance Monitoring System for Squat Exercise by Simple Estimation**

*Mai Tat Chuyen*, Trinh Khanh Ly, Ong Thi Huyen Trang, Nguyen Trung Phu, Nguyen Thanh Hai, Nguyen Thuy Anh, Tran Duc Tan, Nguyen Phan Kien\* - [Springer Nature](#) [🔗](#)

*Proposed a lightweight algorithm using MediaPipe for real-time human pose extraction and joint-angle estimation in motion analysis applications.*

## Experience

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### Research Assistant

*Biomedical Signal and Image Analysis Laboratory (BSI), HUST*

*Sep 2022 – Sep 2025*

- Conducted research on **AI-based breast ultrasound image analysis**, focusing on tumor segmentation and lesion classification.
- Explored multimodal & multitask learning techniques to enhance learning performance.
- Trained large-scale vision models on NVIDIA RTX 3090 GPU with optimized CUDA pipelines for efficient performance.
- Deployed a breast cancer segmentation model as a web-based demo, integrating a Python-based backend inference API with a web frontend interface.
- Deployed ECG classification model on STM32 MCU for real-time arrhythmia detection
- *Tools: PyTorch, Python, Deep Learning, Image Segmentation, Model Optimization, Scientific Writing.*

### Software Engineer Intern

*Motion Analysis Lab, Vinmec International Hospital*

*Sep 2024 – Apr 2025*

- Developed a Python-based **motion analysis and visualization system** integrating data from multiple camera-based measurement tools; **currently used by doctors (Vinmec)** for clinical evaluation.
- **Project code** and description details available at: [Project Page](#) 
- *Tools: Python, Data Processing, Motion Analysis, GUI Development.*

## Skills

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- **Programming:** Python, C/C++
- **AI & ML Frameworks:** PyTorch, TensorFlow, Scikit-learn
- **Data Processing & Visualization:** NumPy, Pandas, Matplotlib
- **Tools & Systems:** Git, VS Code, Jupyter, LaTeX, Linux, Windows, CUDA
- **Language:** English

## Computer Vision and Deep Learning Expertise

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- **Deep Learning:** CNNs, UNet, Transformer, Mamba, RNNs, GNNs
- **Image Processing:** Edge Detection, Denoising, Feature Extraction, Filtering, and Enhancement
- **Generative Vision Models:** GAN, Diffusion Models

## Online Certifications

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- [Google Advanced Data Analytics Professional Certificate](#)  – [Coursera](#) 
- [Machine Learning Specialization](#)  – [Coursera](#) 
- [AI for Medicine Specialization](#)  – [Coursera](#) 

## Professional Activities

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- **Reviewer**, IEEE CIBCB 2025 – 22nd IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology.

## Referees

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Assoc. Prof. Duc-Hau Le, School of Information and Communication Technology, HUST

- [Google Scholar](#)  • Email: hauldhut@gmail.com

Dr. Viet-Dung Nguyen, School of Electrical and Electronics Engineering, HUST

- [Google Scholar](#)  • Email: dung.nguyenviet1@hust.edu.vn